HDOC Day-14 — Menu-Driven Program Using Switch-Case

Problem Statement

Write a menu-driven C program using **switch-case** to input a positive integer and perform one of the following operations:

1. Check whether all digits of the number are unique.

   Algorithm

1. Start.
2. Input a positive integer num.
3. Initialize an array freq[10] with all values as 0.
4. Store num in temp.
5. Repeat while temp > 0:
   - Find the last digit: digit = temp % 10.
   - Check freq[digit].
   - If it is already 1, the number has a repeated digit.
   - Otherwise, set freq[digit] = 1.
   - Remove the last digit: temp = temp / 10.
6. If a repeated digit is found, display **"Not Unique"**.
7. Otherwise, display **"Unique"**.
8. Stop.

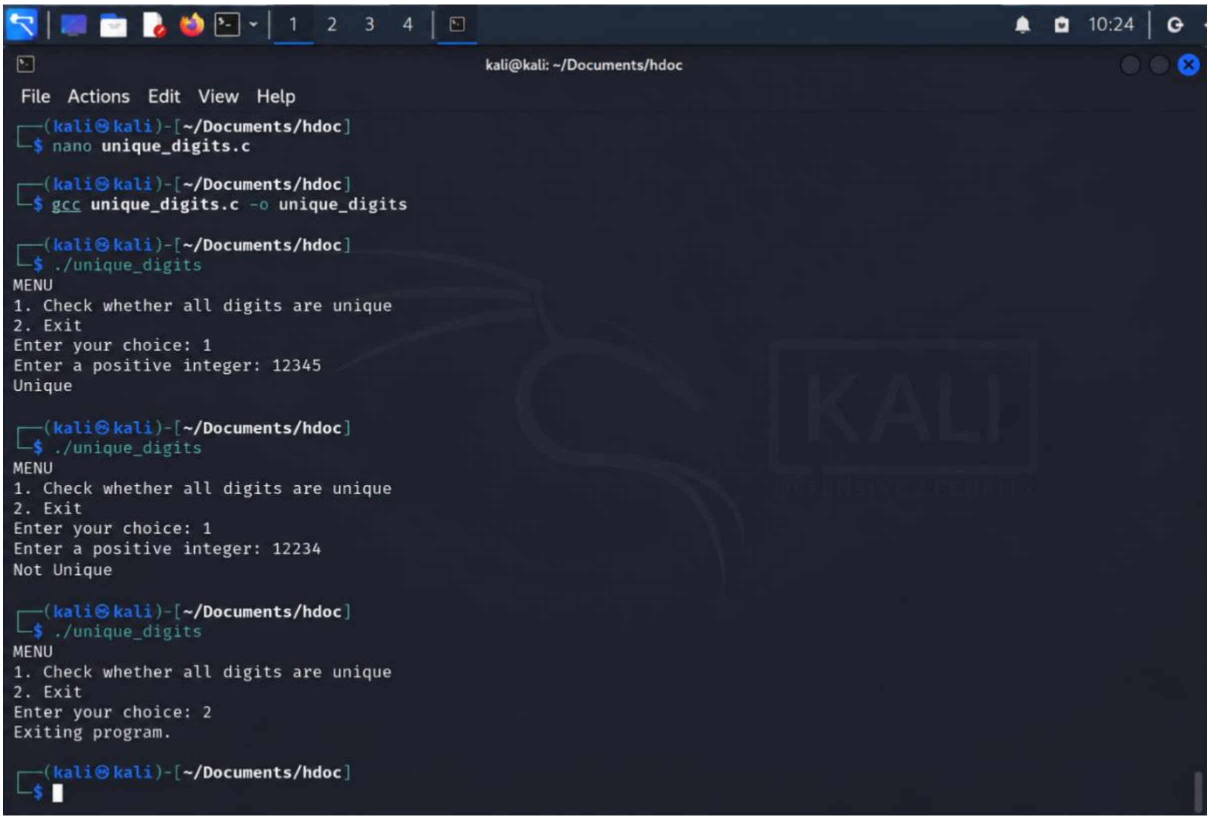
Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	12345	Unique	Unique	Pass
2	12234	Not Unique	Not Unique	Pass
3	987654	Unique	Unique	Pass
4	112233	Not Unique	Not Unique	Pass
5	56789	Unique	Unique	Pass

C Program

```c
#include <stdio.h>

int main()
{
   int num, temp, digit, i;
   int freq[10] = {0};
   int unique = 1;

   printf("Enter a positive integer: ");
   scanf("%d", &num);

   temp = num;

   while (temp > 0)
   {
      digit = temp % 10;

      if (freq[digit] == 1)
      {
         unique = 0;
         break;
      }

      freq[digit] = 1;
      temp = temp / 10;
   }

   if (unique == 1)
```

```
        printf("Unique\n");
    else
        printf("Not Unique\n");

    return 0;
}
```



Result: The program successfully checks whether all digits of the entered positive integer are unique using a switch case menu